Residential Demographic Multipliers for Impact Modeling

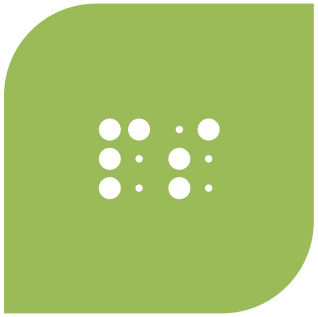
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Acknowledgement
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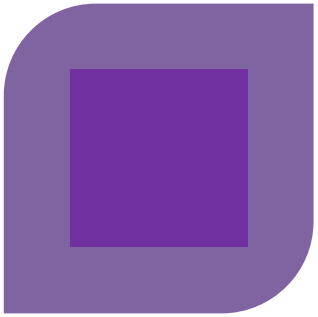
What Are Residential Demographic Multipliers?



SHOW THE AVERAGE NUMBER AND PROFILE OF PEOPLE IN DIFFERENT HOUSING CATEGORIES.



EXAMPLE 3-BEDROOM DETACHED HOME = 2.7 PEOPLE, 0.4 SCHOOL-AGE CHILDREN.



DISTINCTIONS BY HOUSING TYPE, SIZE, TENURE, HOUSING VALUE, AND HOUSING AGE.

Many Critical Uses of Demographic Multipliers

1. Fiscal Impact Analysis (FIA): Estimate municipal/school costs and revenues.

2. School Enrollment Projections: Forecast children from new residential development.

3. Impact Fees: Allocate proportional infrastructure costs (e.g., new school capacity).

4. Public Staffing Analysis: Assess staffing needs (e.g., teachers)

5. Age-Specific Needs: Understand demand for youth/senior services.

6. Market Studies: Evaluate potential buyers/renters and household profiles.

7. Public Policy: Analyze fiscal impact of policies (e.g., climate-related buyouts).

Sources for Demographic Multipliers (DMs)

"Rules of Thumb"– often incorrect

Local survey– (useful, but challenging)

Census/American Community Survey (ACS)– optimal approach

- Public Use Microdata Sample (PUMS)– "raw data" of Census/ACS

The latest Rutgers study (started 1970s) quantifies 50 state DMs from the 2017-2021 ACS (PUMS)

50 State Rutgers Demographic Multipliers (DM) from 2017-2021 ACS

Multipliers comprise:

Household size (HS) – Total persons per housing unit

Age distribution of household members – 0-4, 5-17, 18-34, 35-44, 45-54, 55-64, 65-74, 75+

Total school-age children (SAC)

Total public school-age children (PSAC)*

*SAC who attend public school

SAC and PSAC by grade group – (K-5, 6-8, 9-12)

Demographic statistics (e.g., confidence interval)

50 State Rutgers Demographic Multipliers (DMs) from 2017-2021 ACS **DM DATA FIELDS (Continued)**

Multipliers differentiated by



Rutgers Demographic Multipliers (DM) Data Tools (2017-2021 ACS)

• DMs available in THREE FORMATS for user flexibility:

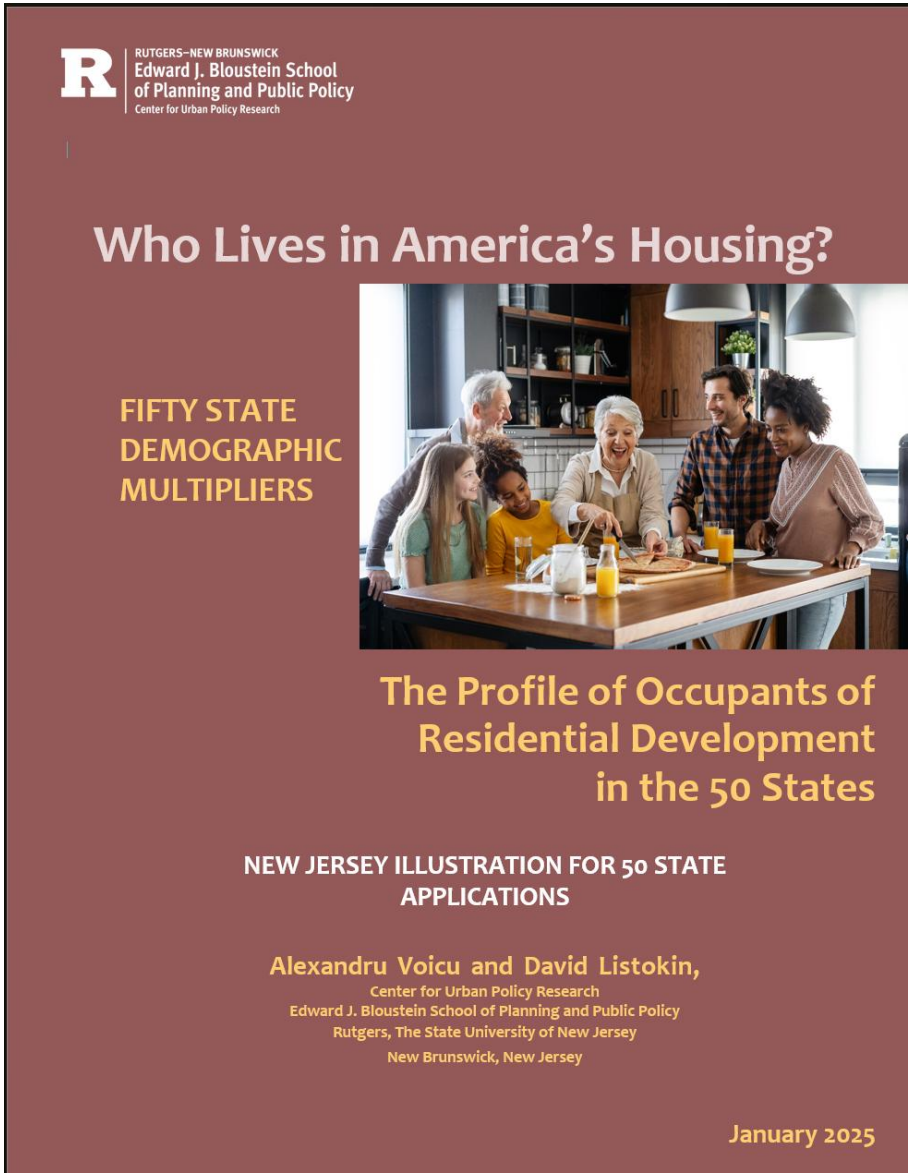
1. State Tables – DMs for each of the 50 States.

2. Online Database – DMs in an interactive tool.

3. Monograph – Comprehensive DMs guide with background, methodology, and a New Jersey case study (for 50 state application)

The monograph is ideal for understanding context and methodological depth, while the state tables and the database are best for rapid access to key DM information.

Rutgers Demographic Multiplier Monograph



Example DM State Table-- New Jersey Excerpt (Excel Tables)

Newer Housing Units

TABLE II-A-1. STATEWIDE NEW JERSEY. TOTAL PERSONS AND PERSONS BY AGE (Newer housing units built 2000-2021, from 2017-2021 ACS)

STRUCTURE TYPE/ BEDROOMS VALUE/TENURE		TOTAL PERSONS	AGE 0-4	5-17	18-34	35-44	45-54	55-64	65-74	75+
Single-Family Detached (Own/Rent), 2 BR										
All Values		1.803	0.035	0.067	0.159	0.063	0.116	0.247	0.560	0.557
Below Median	$310,000	1.865	0.062	0.114	0.256	0.096	0.163	0.228	0.450	0.497
Above Median	$310,000	1.737	0.005	0.016	0.056	0.029	0.065	0.268	0.677	0.621
Single-Family Detached (Own/Rent), 3 BR										
All Values		2.687	0.137	0.428	0.435	0.314	0.408	0.372	0.379	0.214
Below Median	$351,000	2.766	0.165	0.467	0.545	0.371	0.415	0.342	0.291	0.169
Above Median	$351,000	2.605	0.108	0.387	0.321	0.255	0.400	0.404	0.470	0.260
Single-Family Detached (Own/Rent), 4-5 BR										
All Values		3.600	0.219	0.945	0.528	0.515	0.709	0.428	0.173	0.083
Below Median	$553,000	3.568	0.223	0.952	0.548	0.525	0.662	0.398	0.168	0.092
Above Median	$553,000	3.634	0.215	0.937	0.506	0.505	0.759	0.460	0.178	0.074
Single-Family Attached (Own/Rent), 2 BR										
All Values		2.183	0.175	0.289	0.462	0.373	0.225	0.236	0.240	0.182
Below Median	$289,000	2.160	0.136	0.332	0.568	0.313	0.263	0.253	0.136	0.158
Above Median	$289,000	2.207	0.217	0.243	0.348	0.437	0.184	0.217	0.352	0.208
Single-Family Attached (Own/Rent), 3 BR										
All Values		2.830	0.225	0.557	0.583	0.471	0.359	0.329	0.200	0.106
Below Median	$325,000	2.918	0.247	0.633	0.770	0.426	0.354	0.260	0.148	0.080
Above Median	$325,000	2.740	0.202	0.479	0.387	0.519	0.363	0.401	0.255	0.134
2-4 Units (Own/Rent), 0-1 BR										
All Values		1.814	0.122	0.135	0.643	0.310	0.239	0.169	0.129	0.068
Below Median	$152,000	1.638	0.083	0.091	0.457	0.229	0.283	0.177	0.219	0.100
Above Median	$152,000	1.998	0.162	0.181	0.838	0.394	0.192	0.161	0.035	0.035
2-4 Units (Own/Rent), 2 BR										
All Values		2.851	0.297	0.535	0.842	0.466	0.310	0.206	0.134	0.062
Below Median	$185,000	2.821	0.293	0.611	0.778	0.465	0.320	0.194	0.113	0.047
Above Median	$185,000	2.883	0.301	0.456	0.908	0.467	0.300	0.218	0.155	0.077
2-4 Units (Own/Rent), 3 BR										
All Values		3.712	0.372	0.781	0.985	0.598	0.425	0.339	0.158	0.054
Below Median	$231,000	3.866	0.430	0.883	1.027	0.624	0.421	0.322	0.110	0.049
Above Median	$231,000	3.554	0.312	0.676	0.942	0.571	0.430	0.357	0.207	0.060

TABLE IIA1 | TABLE IIA2 | TABLE IIA3 | TABLE IIA4 | TABLE IIA5 | TABLE IIA6 | +

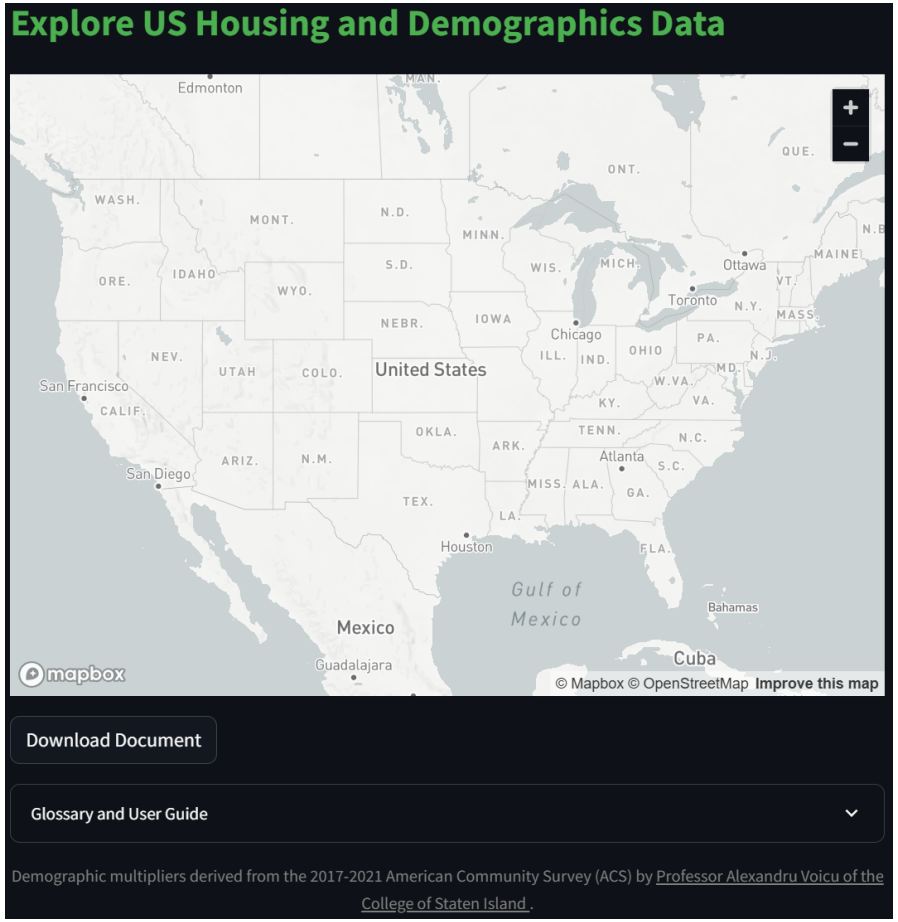
All Housing Units

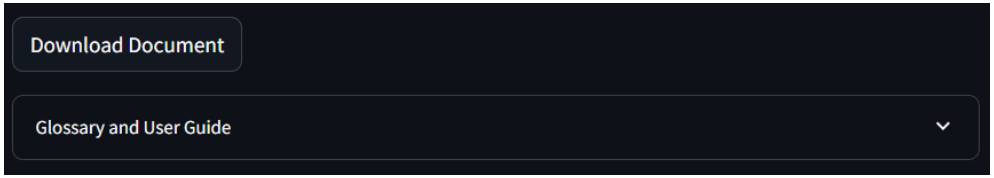
TABLE II-B-1. STATEWIDE NEW JERSEY. TOTAL PERSONS AND PERSONS BY AGE (All housing units, from 2017-2021 ACS)

STRUCTURE TYPE/ BEDROOMS VALUE/TENURE		TOTAL PERSONS	AGE 0-4	5-17	18-34	35-44	45-54	55-64	65-74	75+
Single-Family Detached (Own/Rent), 2 BR										
All Values		2.022	0.075	0.178	0.299	0.197	0.259	0.358	0.344	0.312
First Tercile	Below $193,000	1.988	0.085	0.188	0.328	0.212	0.268	0.350	0.289	0.269
Second Tercile	$193,000 to $307,000	2.010	0.077	0.179	0.302	0.206	0.245	0.350	0.328	0.324
Third Tercile	Above $307,000	2.067	0.064	0.166	0.269	0.174	0.265	0.374	0.413	0.340
Single-Family Detached (Own/Rent), 3 BR										
All Values		2.757	0.144	0.425	0.502	0.342	0.400	0.433	0.310	0.200
First Tercile	Below $260,000	2.754	0.151	0.451	0.574	0.350	0.367	0.396	0.278	0.188
Second Tercile	$260,000 to $398,000	2.762	0.145	0.401	0.511	0.354	0.407	0.440	0.303	0.202
Third Tercile	Above $398,000	2.754	0.138	0.422	0.429	0.324	0.425	0.461	0.345	0.209
Single-Family Detached (Own/Rent), 4-5 BR										
All Values		3.369	0.186	0.683	0.604	0.412	0.531	0.501	0.290	0.159
First Tercile	Below $371,000	3.432	0.198	0.653	0.751	0.406	0.483	0.472	0.291	0.179
Second Tercile	$371,000 to $595,000	3.333	0.178	0.651	0.598	0.408	0.531	0.505	0.301	0.160
Third Tercile	Above $595,000	3.344	0.184	0.744	0.473	0.422	0.577	0.525	0.279	0.140
Single-Family Attached (Own/Rent), 2 BR										
All Values		2.115	0.117	0.274	0.396	0.310	0.264	0.298	0.255	0.201
First Tercile	Below $173,000	2.103	0.135	0.299	0.445	0.280	0.255	0.281	0.209	0.199
Second Tercile	$173,000 to $286,000	2.195	0.117	0.315	0.456	0.355	0.256	0.294	0.228	0.174
Third Tercile	Above $286,000	2.048	0.101	0.211	0.291	0.295	0.280	0.318	0.326	0.227
Single-Family Attached (Own/Rent), 3 BR										
All Values		2.899	0.192	0.564	0.600	0.444	0.387	0.358	0.235	0.119
First Tercile	Below $193,000	3.026	0.195	0.662	0.724	0.436	0.353	0.333	0.207	0.115
Second Tercile	$193,000 to $325,000	2.973	0.225	0.589	0.675	0.462	0.422	0.331	0.188	0.083
Third Tercile	Above $325,000	2.709	0.159	0.447	0.410	0.434	0.385	0.407	0.307	0.159
2-4 Units (Own/Rent), 0-1 BR										
All Values		1.679	0.086	0.135	0.493	0.254	0.215	0.236	0.159	0.102
First Tercile	Below $121,000	1.426	0.054	0.075	0.307	0.167	0.203	0.277	0.208	0.134
Second Tercile	$121,000 to $161,000	1.722	0.094	0.144	0.580	0.286	0.237	0.210	0.113	0.058
Third Tercile	Above $161,000	1.885	0.110	0.183	0.590	0.309	0.205	0.220	0.155	0.112
2-4 Units (Own/Rent), 2 BR										
All Values		2.630	0.200	0.446	0.661	0.378	0.339	0.310	0.168	0.127

TABLE IIB1 | TABLE IIB2 | TABLE IIB3 | TABLE IIB4 | TABLE IIB5 | TABLE IIB6 | +

Demographic Multipliers Database

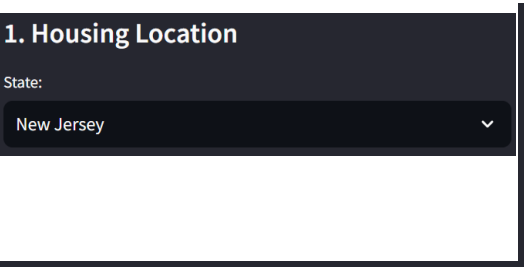


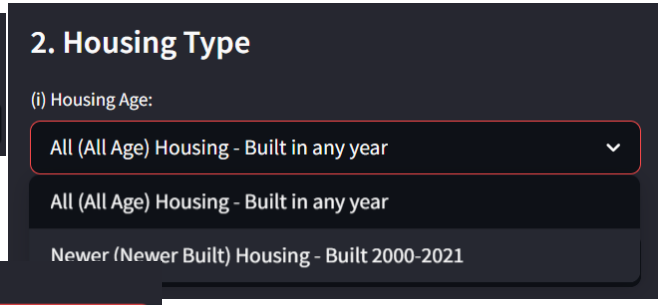


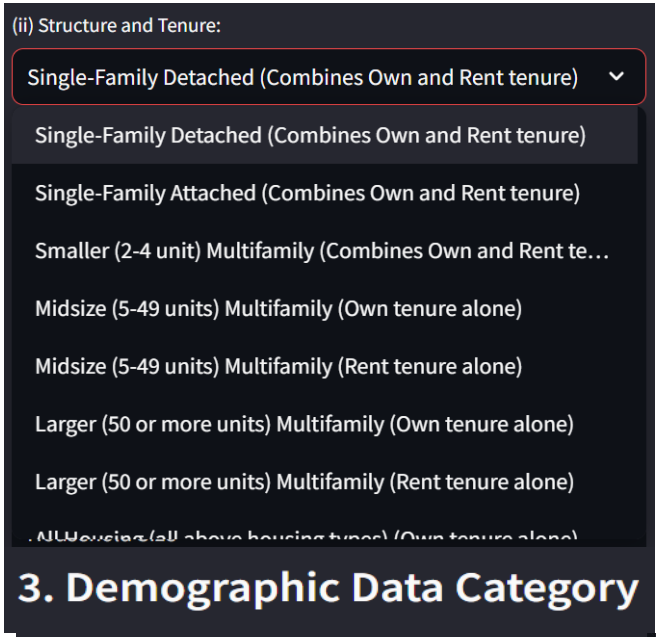
Detailed Overview, user guide, and definitions are presented.

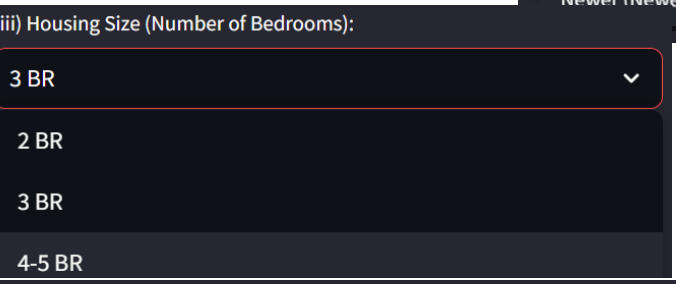
https://demographics-rutgers.streamlit.app/

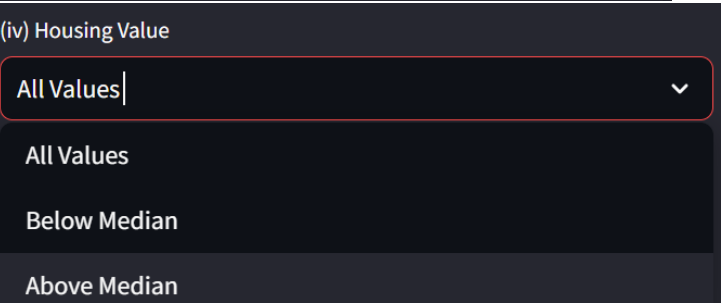
Demographic Multipliers Database

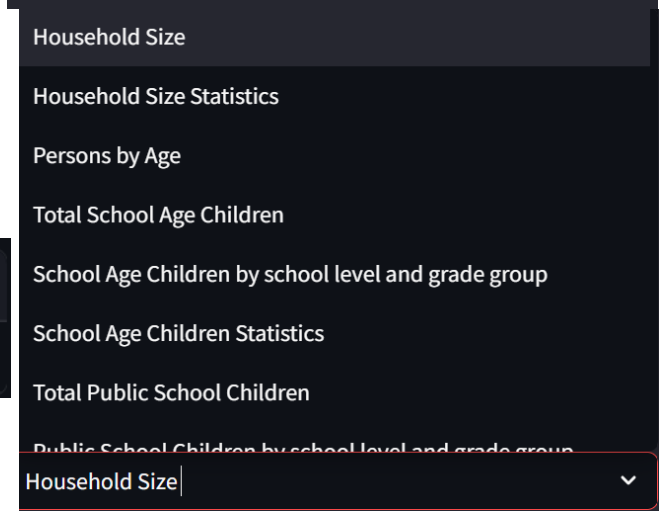












Outputs:
Household Size = **2.6053**

	Structure	VALUE_TENURE	value_range	PERSONS
5	Single-Family Detached (Own/Rent), 3 BR	Above Median	351,000	2.6053

Rutgers Demographic Multiplier Database

URL: https://demographics-rutgers.streamlit.app/

Interactive filters for state, and housing unit age, type, size (bedrooms), value and tenure.

Example 1: New Jersey, Newer housing, 3-bedroom, single-family detached (SFD), own/rent, above median value. Household Size and Public-School Children

Outputs: Household Size = **2.6053, PSC= 0.3174,** Public School Children

Example 2: New Jersey, All age housing, 2 BR, 50+unit building, own, all values. Household size and Public-School Children

Outputs: Household Size **1.8457**

Public School Children: **0.0877**

Key Demographic Multiplier (DM) Takeaways

DMs provide key information for many critical applications.

Rutgers provides DMs for all 50 states from 2017-2021 ACS

Rutgers DMs are in tables, database, and monograph forms.

Future Demographic Multipliers (DM) Research



Quantify DMs for affordable housing (low- and moderate-income)



Refine DMs (from American Housing Survey) by:

Building height (multifamily low- and high-rise)

Age-restricted homes

Vacation/second homes/subsidized units



Refine nonresidential multipliers (workers per 1,000 sq. ft.)



Expand statistical analyses

National Science Foundation

RUTGERS–NEW BRUNSWICK
Edward J. Bloustein School of Planning and Public Policy
Center for Urban Policy Research

MACH
Megalopolitan Coastal Transformation Hub

NSF

COPE
COASTLINES AND PEOPLE

App URL: https://demographics-rutgers.streamlit.app/
Please share your thoughts!

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Umer Farooq (uf12@scarletmail.rutgers.edu)

Demographic and Fiscal Modeling for Environmental Planning: New tools from NSF-funded projects
Gov. James J. Florio Special Events Forum
33 Livingston Avenue, New Brunswick, NJ
April 22, 2025

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